

packages/graph-core/src/change-classifier.ts

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Overview

A new file `packages/graph-core/src/change-classifier.ts` introduces a comprehensive update-decision engine for graph rebuilds, replacing the previous ad-hoc logic.

Key changes

- **New `UpdateDecision` interface** (lines 4-10) defines `action`, `filesToReanalyze`, `rerunArchitecture`, `rerunTour`, and `reason`.
- **`classifyUpdate` function** (lines 21-86) now evaluates structural changes, file counts, and directory shifts to return an `UpdateDecision`.
- **Directory-aware detection** via `detectDirectoryChanges` (lines 94-113) and `topDirectory` (lines 119-124) checks for new or removed top-level directories.
- **Human-readable summary** produced by `summarizeChanges` (lines 129-143) for logging and diagnostics.
- Added imports: `dirname` from `node:path` and a type import for `ChangeAnalysis` (lines 1-2).

Impact

- **Correctness:** Explicit decision matrix reduces ambiguous rebuild triggers; thresholds for full vs. partial updates are now deterministic.
- **Maintainability:** Centralized decision logic in a single module simplifies future rule tweaks and unit testing.
- **Observability:** `reason` strings provide clear audit trails for why a rebuild was chosen.
- **Compatibility:** No API changes to existing callers; the new module is self-contained and only adds exports.

Risks & follow-ups

- **Regression risk:** Verify that existing build pipelines still trigger the expected actions; unit tests for `classifyUpdate` should cover edge cases (e.g., >30 structural files).
- **Performance:** `detectDirectoryChanges` iterates over all known files; benchmark on large projects to ensure no slowdown.
- **Type safety:** Ensure `ChangeAnalysis` type aligns with callers; missing fields could cause runtime errors.
- **Documentation:** Update README or internal docs to explain the new decision matrix and thresholds.